Instance-Level Multitask Learning for 3-D Building Extraction From Monocular Off-Nadir Satellite Sensor Imagery

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***Abstract*— Extracting 3-D building information from monocular satellite sensor imagery remains a formidable challenge in the field of remote sensing. Multitask frameworks based on deep learning, which are particularly for simultaneously predicting 2-D building outlines and their respective heights using orthorectified satellite imagery, have shown promise in addressing this challenge. Height estimation is notably complex due to the absence of explicit height indicators, limited interaction between semantic-height features, and inadequate representation of building relationships. Moreover, the availability of data sources is a limiting factor for broader application. To overcome these issues, this study introduces an innovative instance-level multitask learning model (named BDH-Net) that leverages off-nadir perspectives and roof-to-footprint offset vectors to enhance modeling. This model comprises four key components: a pixel-wise feature extraction image encoder–decoder, a query Transformer decoder, a multitask decoder, and a height decoder that employs intrainstance and interinstance attention for precise building height estimation. Additionally, we pioneer the use of Google Earth imagery to construct an off-nadir satellite dataset with roof-to-footprint offset vectors specifically designed for building instance segmentation and height prediction, known as the Building Height (BDH) dataset. Comprehensive experiments demonstrate that the proposed BDH-Net significantly improves the accuracy of monocular 3-D building data extraction by integrating roof-to-footprint offset vectors and leveraging context specific to each building instance. With the extensive coverage and regular updates of Google Earth imagery, BDH-Net holds substantial potential for wide-ranging and long-term applications. The source code of the proposed BDH-Net and the BDH dataset is publicly available at https://github.com/wishx98/BDHNet**

***Index Terms*— 3-D building information extraction, building height estimation, deep learning, instance segmentation, monocular off-nadir satellite, multitask learning, remote sensing.**

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I. Introduction

URBAN landscapes are characterized by their diverse building structures, fulfilling residential, commercial, and cultural roles. The extraction of precise 3-D information from these structures is essential for supporting a wide range of applications, including delineating urban functional zones [1], [2], [3], enhancing scene comprehension [4], optimizing structural layouts [5], [6], aiding in real estate assessment [7], monitoring environmental trends [8], [9], [10], and facilitating disaster response [11]. Despite its critical importance for urban modeling and monitoring, accurately estimating building heights presents significant challenges, including technological limitations and the high costs associated with data collection and processing.

The quest for accurate 3-D building information has catalyzed the development of diverse methods. Prominently, airborne light detection and ranging (LiDAR) technology excels in urban 3-D reconstructions, offering high precision by isolating and capturing detailed 3-D data of buildings, often with submeter accuracy. Similarly, inspired by advancements in stereoscopic technology, techniques utilizing multiple stereoscopic image pairs have advanced considerably, enhancing 3-D urban modeling. This progress is supported by satellites like the WorldView series, GEOEye, Pleiades, ALOS, GaoFen-7, and ZiYuan-3, which supply stereoscopic images that enable the generation of accurate digital elevation models (DEMs) and digital surface models (DSMs). Despite these advances, the integration of LiDAR and multiview optical imaging technologies encounters significant hurdles, mostly related to the economic and logistical challenges inherent in urban data collection. The high costs associated with acquiring, processing, and maintaining up-to-date high-resolution urban datasets result in a significant gap in 3-D building data availability. Moreover, techniques that rely on synthetic aperture radar (SAR) and the fusion of data from

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multiple sources face their own set of limitations, highlighting the complexity and challenges of achieving comprehensive urban 3-D reconstructions.

The emergence of deep learning and substantial progress in monocular depth estimation, particularly within the autonomous driving field, have provided new possibilities for reconstructing 3-D scenes from single 2-D images. This progress has spurred the development of innovative approaches for estimating building heights from monocular orthorectified images [12], [13], [14], [15], highlighting the potential of convolutional neural networks (CNNs) and deep learning technologies. Notably, techniques such as [14] have demonstrated the efficacy of CNNs in extracting urban height maps directly from RGB remote sensing images, representing a notable advancement in precision and efficiency, while techniques in [12] have enhanced extraction precision through deep residual learning and skip connections.

The recognition of unique height distribution patterns across various object categories has prompted a reevaluation in the extraction of 3-D building information, leading to the adoption of multitask learning frameworks. These frameworks combine monocular height estimation with semantic segmentation to enhance 3-D modeling accuracy. An early example [16] demonstrated how integrating semantic analysis with height prediction could significantly improve accuracy, underscoring the synergistic relationship between recognizing object categories and measuring spatial dimensions. Despite these advancements, some methodologies like IMG2DSM [17] and MSCDN [18] employ separate decoders for varying tasks without adequately modeling the interaction between height and category distribution, a critical aspect for accurate 3-D modeling. Recent studies aim to bridge this gap by fostering interactions between semantic and geometric features, with [19] proposing a model that aggregates semantic and geometric information at an intermediate stage for enhanced feature interaction. Furthermore, Feng et al. [20] introduced a calibration refinement attention module that filters inconsistent features across tasks, refining the accuracy and reliability of 3-D building information extraction.

Nevertheless, the development of these promising methodologies that utilize orthorectified imagery is accompanied by significant challenges. One major challenge is accurately deriving height from monocular images without direct height data, often requiring reliance on heuristics, contextual insights, or assumptions about the scene, which can compromise estimation accuracy. Additionally, the integration between semantic and height information remains a critical area for improvement, as the accuracy of height predictions is closely tied to the precision of category predictions. The current reliance on pixel-level semantic segmentation to inform height estimation sometimes leads to discrepancies in height predictions within buildings, especially when semantic interpretations are inaccurate, indicating ongoing challenges in fully realizing the potential of these advanced methodologies for accurate 3-D urban modeling. Additionally, the availability of large-scale, regularly updated, high-resolution orthorectified imagery is limited, which further restricts the potential application of these methods.

Our research introduces two foundational insights that set our work apart in the urban 3-D modeling domain, as illustrated in Fig. 1. This figure delineates the differences between our innovative method, reliant on off-nadir imagery and instance-level feature extraction, and conventional strategies grounded in orthorectified imagery and pixel-level analysis, thereby underscoring the strengths of our approach.

Our method leverages the unique height cues available in off-nadir imagery to determine building height accurately. The roof-to-footprint offset vector, which refers to the vertical displacement between the roof of a building and its base, becomes increasingly distinguishable and measurable as the off-nadir angle expands. This characteristic serves as a direct proxy for determining building height. Li et al. [21] and Wu et al. [22] have validated the effectiveness of utilizing pixel-level offset vectors for monocular height estimation. Nonetheless, Li et al. [21] faced challenges in their paradigm of separately extracting roof and facade information for offset vector calculation. Specifically, the use of semantic segmentation tasks complicated the precise alignment of facades with roof and footprint edges, crucial for accurately deriving offset vectors, leading to potential misalignment and diminished precision. Moreover, the challenge of extracting a building's footprint from off-nadir images via semantic segmentation becomes apparent, with roof extraction proving to be more reliable. The partial visibility of the footprint complicates the precise and complete determination of a building's geographic location. In response to these challenges observed in pixel-level techniques, our strategy shifts toward the utilization of instance-level features. By emphasizing instance-level data and applying instance-level roof-to-footprint offset vectors, we can conduct direct supervision training on the instance-level roof-to-building offset vector. Furthermore, by shifting the extracted roof along the direction and magnitude of the offset vector in the x- and y-axes, we can accurately obtain the footprint. Therefore, our method not only addresses these issues of precision but also circumvents the reliance on imprecise heuristics.

Building on the foundational insights outlined, our research further distinguishes itself by incorporating instance segmentation as a fundamental component of our methodology. This approach integrates various data dimensions—height, category, boundary information, and architectural characteristics—into a unified analysis framework. Such integration enables the detailed examination of both the relationships within individual building instances and those between different instances, offering a comprehensive view of building 3-D information. This method contrasts with traditional multitask building height prediction approaches that predominantly utilize orthorectified imagery through instance segmentation [23], [24], [25]. The dependency on orthorectified images, due to their lack of direct height indicators, often undermines the accuracy and effectiveness of these techniques. Our instance-based approach, by leveraging off-nadir imagery and instance-level data, effectively overcomes these limitations, showing a notable improvement in precision and reliability for urban 3-D modeling.

In summary, we present a building height network called BDH-Net, a network that embodies a multitask framework that

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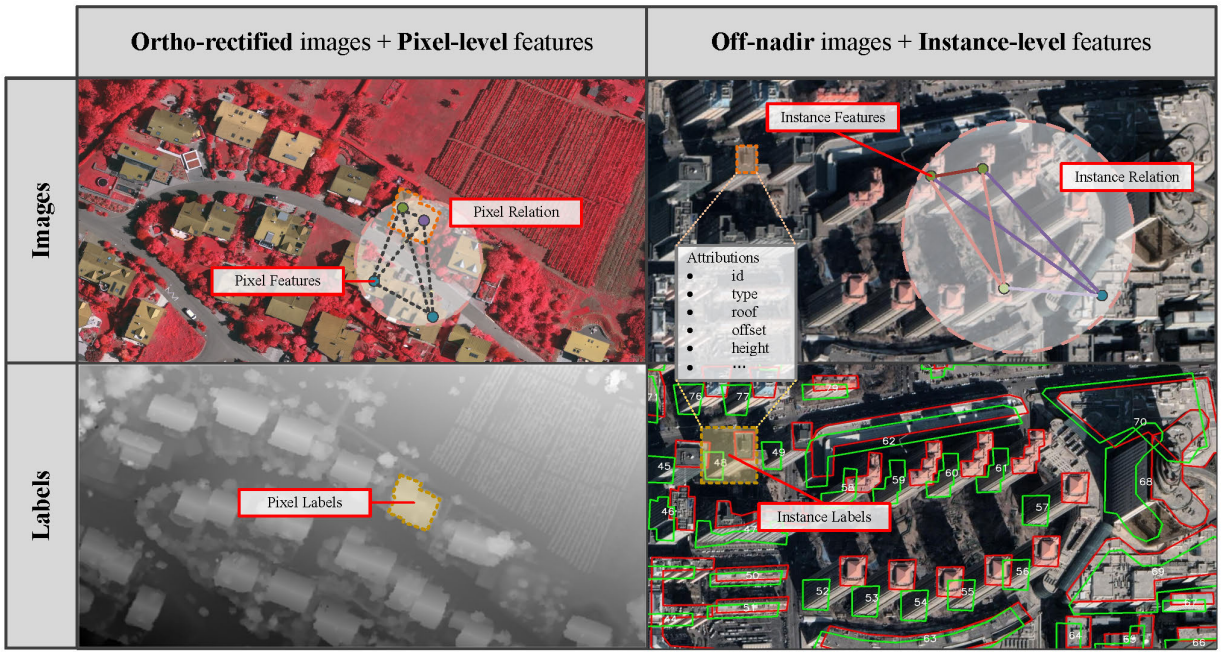


Fig. 1. Comparative display of monocular height estimation techniques at the pixel and instance levels. (Left) Pixel-level characteristics and interactions within orthorectified imagery, each pixel labeled accordingly. (Right) Off-nadir imagery is employed to highlight instance-level features and spatial relationships, with unique labels for each building.

utilizes Google Earth imagery. This framework is designed to leverage the unique perspectives provided by off-nadir imagery and the integration of instance-level data, structured around four foundational components: a multiscale image encoder–decoder network, a query Transformer decoder, a multitask decoder, and a height decoder: 1) the multiscale image encoder–decoder network is tasked with extracting primary pixel-level features across multiple scales; 2) the query Transformer decoder then processes these pixel-level features to extract instance-specific attributes, facilitating the interaction of instance-level features such as category, building boundaries, height, and offset vectors; 3) the multitask decoder is designed to extract key instance attributes, including height, category, boundaries, offset vectors, instance centers, and scale; and 4) the height decoder incorporates image scale features and instance positional information, initially conducting intrainstance feature self-attention calculations to extract representative features that efficiently characterize building instances. Subsequently, the method employs an interinstance attention graph (IAG) to model the relationships between building instances. This attention graph, along with the positioning of instance centers, significantly improves the modeling of spatial relationships among instances. This feature is particularly valuable in scenarios where some buildings within the same district are obscured. It enables the system to rely on height information from unobstructed buildings within the same area, offering a reliable approach to height estimation even in instances of partial occlusion. Finally, an optimized adaptive bin height regressor is employed to robustly extract building heights. Through extensive experiments and the creation of the Building Height (BDH) dataset, our goal is to significantly advance the field of monocular 3-D building extraction and establish a useful benchmark for research in off-nadir 3-D modeling. Benefiting from the widespread availability and regular updates of Google Earth imagery, BDH-Net boasts promising application prospects and the capability for extensive, multitemporal deployments. This framework not only enhances the accuracy and efficiency of 3-D urban modeling but also supports a broad spectrum of applications across different temporal and spatial scales.

The main contributions of this work are as follows.

1) We propose a multitask instance segmentation approach for monocular height estimation leveraging off-nadir images, offering a potential avenue for future research.
2) We introduce techniques for intrainstance feature recalibration and interinstance graph modeling, facilitating the detailed analysis of relationships within and between building instances.
3) We develop a comprehensive off-nadir satellite dataset for 3-D building modeling over the 5th Ring Road region of Beijing, providing a useful benchmark for further research in this area.

This article comprises five additional sections. Section II delves into a literature review. Section III provides insights into the data, while Section IV outlines the methodology. Section V introduces the settings of our experiments. Sections VI and VII present the results and a discussion, respectively. Finally, Section VIII offers a concise summary of the conclusions.

II. Related Works

A. Building 3-D Information Extraction

Traditional methodologies for estimating building heights are primarily centered around six techniques: LiDAR, stereoscopic analysis of multiview optical images, SAR image inference, nadir optical image inference, off-nadir optical image inference, and the complex fusion of multisource data.

LiDAR technology has emerged as a key technology in estimating building heights, utilizing its capacity to directly

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measure the distance between the ground and objects. This method provides high-precision 3-D spatial data, ideal for capturing the vertical structure of urban environments [26], [27]. This technology generates a dense point cloud that encompasses the Earth's surface and various objects, necessitating a sophisticated classification process to distinguish building points for precise height estimation and 3-D modeling [28], [29], [30], [31]. Meanwhile, the analysis of multiview optical images offers another approach for reconstructing detailed 3-D urban models. By examining images captured from various angles, this approach delivers detailed insights into the intricate details of urban forms [32], [33], [34], [35]. However, LiDAR and multiview optical images face the challenge of reconciling the potential of these technologies with the financial and logistical realities of urban data collection. The high costs associated with collecting, processing, and updating high-resolution urban datasets result in a persistent gap for 3-D building information.

SAR technology is notable in remote sensing imagery with its ability to penetrate cloud cover and operate independently of lighting conditions. The utility of SAR in height estimation primarily relies on analyzing radar shadows and layovers, phenomena indicative of structure heights, using complex geometrical structure prediction and matching strategies [36], [37], [38], [39], [40]. Despite its potential, SAR-based building height estimation faces considerable challenges in urban environments, such as complex signal phenomena [39], partial occlusion [41], and the variability in reflection characteristics of building materials [42]. These factors complicate the interpretation of SAR data, especially in densely built areas where radar shadows, layovers, and the double-bounce effect are crucial for determining building heights.

Nadir and off-nadir optical images serve as the primary data source for monocular building height estimation, providing a unique perspective and method for extracting structural dimensions in urban landscapes. While the nadir optical image provides a direct overhead view, which is useful for mapping and horizontal measurements, it lacks the vertical dimension required for accurate altitude determination. Therefore, studies based on nadir optical images usually adopt data-driven methods, using shadow features [43], [44], [45], [46], [47] or multidimensional features represented by models [48], [49] to indirectly infer building heights. On the other hand, off-nadir optical imagery introduces an angled perspective, revealing building facades and significantly reducing occlusion issues encountered in nadir views. This perspective facilitates more direct height estimation through detailed analysis of building facades and shadow geometry. However, the benefits of off-nadir imagery come with the challenge of perspective distortions, necessitating complex physical modeling or a hybrid of physical modeling and semi-data-driven methods for accurate geometric information extraction.

The fusion of multisource data, while aiding in a more comprehensive understanding of urban landscapes, often confronts issues related to data heterogeneity, alignment, and the complex process of effectively integrating different types of data [50]. Though leveraging multisource data for remote sensing research is increasingly popular, the inherent difficulties it presents in estimating building heights are similarly inescapable.

B. Instance Segmentation

Instance segmentation differs from semantic segmentation [51], [52] by assigning semantic labels and individual instance IDs to each pixel. This process provides more detailed information than semantic segmentation, enabling precise identification of object locations, boundaries, and counts [53]. A key development in instance segmentation is the introduction of Mask R-CNN [54], which builds upon the object detection framework of the region CNN (RCNN) [55] series by adopting the region of interest align (ROIAlign) method. Mask R-CNN and its derivatives have significantly impacted the field of remote sensing, supporting a variety of applications, including building recognition [56], tree detection [57], and cloud detection [58].

The integration of Transformers into object detection and instance segmentation represented a notable innovation with the development of DEtection TRansformer (DETR) [59], the first method to apply the Transformer architecture fully to these tasks, directly predicting object boundaries and segmentation. Subsequent innovations like Maskformer [60] and Mask2former [61] treated instance segmentation as a per-pixel classification problem, effectively merging instance and semantic segmentation into a unified model. These advancements highlight the evolution of instance segmentation technologies, improving precision and flexibility in object detection tasks.

Advancements in building instance segmentation have shown significant progress. Some methods continue to utilize the approach of Mask R-CNN, which initially predicts bounding boxes for buildings and then refines these to delineate building contours [62]. Some other researchers treat building instance segmentation as the sequential prediction of multiple vertices, employing recurrent neural networks (RNNs) and their variants to iteratively predict the vertices of buildings [63], [64]. Additionally, some studies first generate a probability map of the buildings and subsequently simplify these data into vector boundaries using contour or corner detection techniques [65], [66].

C. Monocular Height Estimation

The development of monocular height estimation has been significantly influenced by advances in monocular depth estimation, adapted for the specific demands of remote sensing applications. Initial explorations in the field of monocular depth estimation, such as the work in [67], underscored the importance of leveraging both global and local information from various cues within a single image to infer depth relationships, proposing a dual-branch model to extract these distinct feature sets. Further advancements in [68] showed the complementary nature of surface normal prediction and semantic segmentation in enhancing monocular depth estimation, introducing a gradient loss function designed to refine predictions by emphasizing similar local structures and gradient variations in different directions. Building on these

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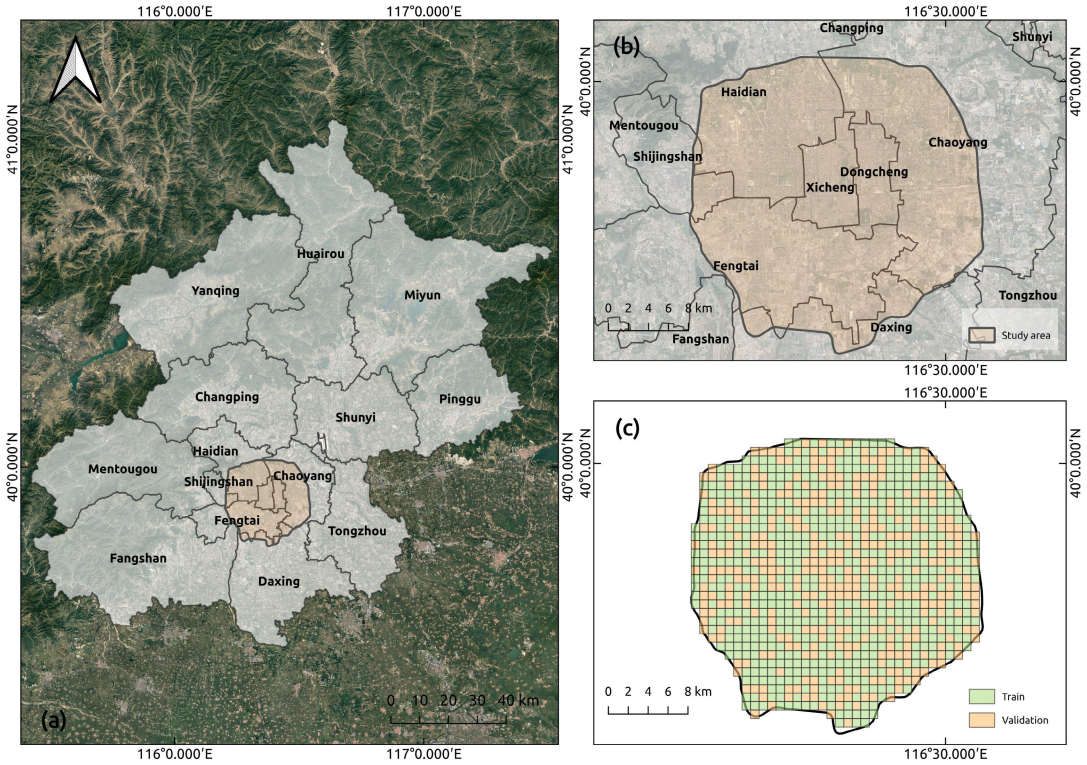


Fig. 2. Geographic overview and data partitioning of the study area. (a) Extensive geographical setting, (b) administrative divisions within the focal study region, and (c) allocation of the BDH dataset for training and validation within the study area.

insights, Fu et al. [69] reconceptualized depth estimation as a classification problem, providing fresh perspectives on tackling this challenge. Bhat et al. [70] extended this approach by combining classification with regression for improved accuracy in depth prediction, marking a key evolution in the field.

Initially, the transition from depth to height estimation in monocular settings utilized orthorectified imagery and relied on standard encoder–decoder architectures for feature extraction, coupled with streamlined height regressors for generating DSMs or estimating building heights [14], [16], [17]. Subsequent research has proposed the use of independent decoders dedicated to distinct tasks, which enables better feature interaction and alignment, as evidenced in studies in [20], [71], [72], and [73]. This strategy allows for early differentiation of features relevant to each task, permitting the selective exchange of features that are mutually beneficial while avoiding the mixing of task-specific characteristics. A key element in this refined approach is the use of feature recalibration and attention mechanisms to optimize the synergy between tasks. For instance, Feng et al. [20] implemented locally dense attention to sift through and refine feature consistency, whereas Cai et al. [71] introduced a multimodal selective-kernel module to enhance the extraction of integrated information from both optical and SAR imagery. These innovations are current research directions in monocular height estimation, showing a detailed understanding of feature dynamics and task-specific requirements that drive the field forward. In addition to pixel-level instance segmentation methods, related studies have also explored the feasibility of instance segmentation for monocular height estimation on orthorectified imagery. Existing research primarily involves adding a height prediction head to the instance segmentation model [23], [24], [25]. Like all pixel-level methods based on orthorectified imagery, these approaches are limited by the orthorectification process, which inherently lacks explicit height cues.

Few studies have investigated monocular height estimation from off-nadir imagery [21], [22], [74]. Most of them have utilized the roof-to-footprint offset vector as a proxy for height and calculated the offset vector through the building facades. However, these methods are still confined to pixel-level instance segmentation approaches, which encounter specific challenges: 1) semantic segmentation can lead to misalignment between the building's roof and footprint edges limiting the precision of offset vector extraction and 2) extracting a building's footprint from off-nadir images using semantic segmentation is challenging. The invisibility of certain areas of the footprint makes it difficult to accurately and completely predict the building's geographical location.

III. Study Areas and Data

A. Study Areas

Beijing, nestled between 39.4°N to 41.6°N and 115.7°E to 117.4°E, has an average elevation of 43.5 m above sea level. The heart of our study encompasses Beijing's core urban zones, specifically the administrative precincts of Dongcheng, Xicheng, Haidian, Chaoyang, Shijingshan, Fengtai, and Daxing. These districts, circumscribed by the 5th Ring Road of Beijing, serve as our primary area of interest. Despite these areas making up only about 10% of Beijing's total land area, their influence on the city's economy and demographics is significant. They contribute to more than 70% of Beijing's

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Fig. 3. Visual representation of the BDH dataset. (a) and (c) Raw satellite imagery from Google Earth. (b) and (d) Annotated details with building roofs marked in red and building footprints outlined in green.

Gross Domestic Product (GDP) and are home to over half of the city's population. From a construction perspective, these regions account for more than 40% of Beijing's total residential construction area.[1] A comprehensive visualization of the study locale is provided in Fig. 2(a) and (b).

B. Remote Sensing Data Collection and Processing

To meet the methodological prerequisites of acquiring fine-resolution imagery within a defined temporal frame, we sourced Google Earth imagery at N19-level (https://earth.google.com), dated January 18, 2023. This imagery, with a ground sample distance of 0.54 m, encompasses the RGB spectrum, spanning the entire study area. A challenge arose due to the heterogeneous sensors used in capturing the imagery, each with unique angles and intervals, as evidenced in Fig. 3. This intricacy impacted the subsequent methodological design. The imagery was then segmented into 2048 $\times$ 2048 pixel tiles with no overlap.

C. Building Dataset and Preprocessing

This research relies heavily on a meticulously curated building dataset, encompassing two critical aspects: building height data and roof-to-footprint offset vector data.

1) Building Height Data: The building height data used in this research were sourced from AMap, a renowned Chinese cartographic service provider known for its comprehensive vector data. The dataset, acquired from the year 2019, provides building height information derived using a standard floor-to-height conversion factor of 3 m/floor, a common practice in the absence of direct height measurements. While this assumption is widely accepted, variations in floor-to-height coefficients depending on building type and location, such as those reported in [75], were carefully considered to improve data reliability.

To address these potential variations, we implemented a rigorous validation process. First, we cross-referenced the AMap dataset with building data from Lianjia, a leading Chinese real estate platform, which provided detailed floor level and height information primarily for residential and mid-rise buildings. For commercial office buildings and other specific structures, where Lianjia's data were less applicable, we manually consulted official websites and property listing platforms to obtain accurate height and floor number information. In instances where the data appeared erroneous or no reference data was available, we conducted on-site measurements using the SNDWAY SW-800B infrared rangefinder, which offers a distance measurement accuracy of ± 1.

2) Building Roof-to-Footprint Offset Vector Data: The creation of this dataset involved a team of ten experts in geographic information and cartography. The workflow for delineating these features began with the careful delineation of building roofs, chosen for their ability to reduce occlusion and enhance boundary accuracy. Subsequently, building footprints were generated by translating the roof shapes to obtain footprints mirroring the roof contours. Finally, the offset vectors, central to our research, were meticulously computed.

In the preprocessing phase, two crucial tasks were executed. First, we established correspondences between the polygons in the AMap dataset and the roof-to-footprint dataset. This was primarily done using intersection-over-union (IoU) metrics and an assessment of shape similarity (referred to as "sm"). The shape similarity was defined as follows:

$$\mathrm{sm}(p1, p2) = \frac{1.0}{1.0 + |\mathrm{ar}(p1) - \mathrm{ar}(p2)|} \tag{1}$$

$$\mathrm{ar}(p) = \frac{\mathrm{bound}(p).\mathrm{width}}{\mathrm{bound}(p).\mathrm{height}} \tag{2}$$

[1]Data source: Beijing Regional Statistical Yearbook 2022.

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TABLE I
DETAILED OVERVIEW OF THE BDH DATASET

City	District	Area (km^2)	Image Samples (Total/Train/Val)	Instance Samples (Total/Train/Val)
Beijing	Dongcheng	41.8	62/ 34/ 28	6225/ 3290/ 2935
	Xicheng	50.2	68/ 31/ 37	6174/ 2752/ 3422
	Haidian	148.3	221/ 130/ 91	19836/ 11934/ 7902
	Chaoyang	193.3	260/ 154/ 106	16451/ 9784/ 6667
	Shijingshan	16.6	32/ 20/ 12	2112/ 1095/ 1017
	Fengtai	173.3	240/ 159/ 81	14013/ 10193/ 3820
	Daxing	43.8	48/ 30/ 18	2443/ 1716/ 727
Total	5th Ring Road region	667.4	931/ 558/ 373	67,254/ 40,764/ 26,490

Note: All image samples are of dimension 2048×2048 pixels. Furthermore, each instance sample comprises both a roof and its corresponding footprint.

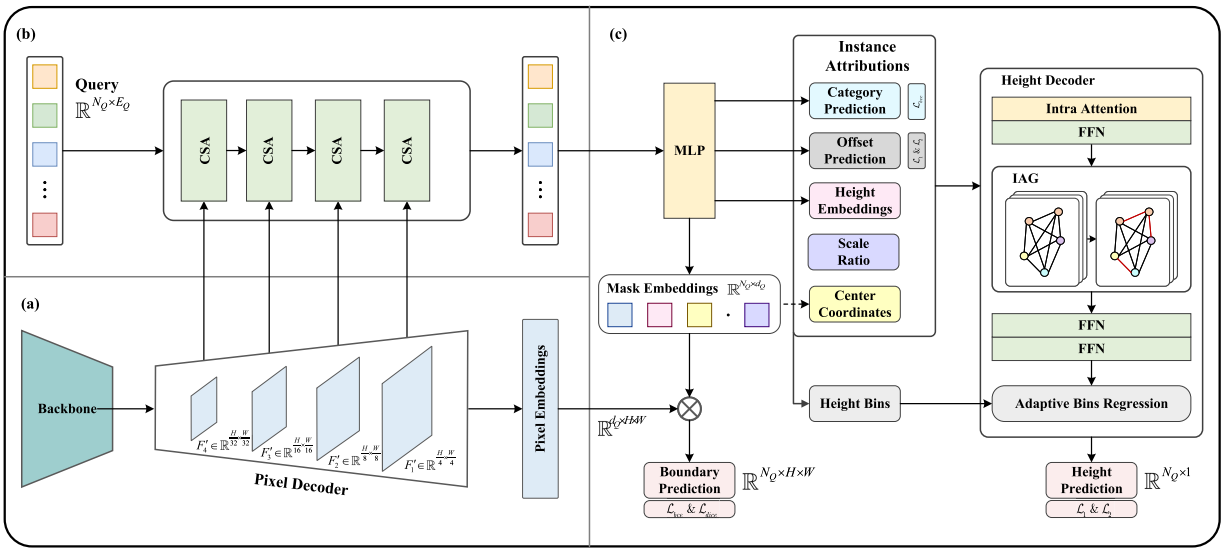


Fig. 4. Complete structural framework of the BDH-Net, showcasing the integration of various components within the network. (a) Encoder–decoder module. (b) Query Transformer decoder. (c) Multitask decoder.

where p represents a building instance and the term bound(·) denotes obtaining the minimum bounding rectangle for the building instance.

Furthermore, we utilized the offset vectors from the roof-to-footprint dataset in conjunction with the height data obtained from the AMap dataset to conduct a preliminary assessment of height correctness. Any height anomalies discovered during this evaluation underwent a meticulous process of manual validation and rectification. The corrected height data were then seamlessly integrated into the roof-to-footprint dataset, resulting in a dataset of good quality and integrity.

D. BDH Dataset

The BDH dataset assumes a central role in our research endeavor, constituting the foundation upon which the core objectives of our study are built. This dataset, carefully curated, offers precise insights into individual building instances and their corresponding heights. A comprehensive overview of the BDH dataset can be found in Table I. The coordinate system used for these data is the WGS 84/Pseudo-Mercator (EPSG:3857). This dataset comprises a total of 931 image-level samples, each boasting dimensions of 2048 × 2048 pixels. In addition, it encompasses a large number of over 67 000 instance-level samples. For the purpose of training and validation, these samples are allocated in a 60:40 ratio, respectively. Notably, the geographical coverage of this dataset spans a sprawling area of 667.4 km^2 within the Beijing region, confined by the boundaries of the city's 5th Ring Road. This dataset is characterized by its unique role as one of the first off-nadir satellite datasets specifically curated for the nuanced demands of building instance segmentation and height estimation. It serves as a significant resource for researchers in the field.

IV. BDH-NET FRAMEWORK

This section introduces the BDH-Net framework and the loss functions used for supervising and balancing multiple tasks.

BDH-Net is a multitask learning framework inspired by Mask2Former [61], simultaneously predicting building instance segmentation and height estimation. It comprises four key components: 1) a Swin Transformer-based image encoder–decoder; 2) a query decoder; 3) a multitask decoder; and 4) a height decoder. The overall architecture of BDH-Net is illustrated in Fig. 4. The detailed descriptions of each component are provided in Sections IV-A–IV-D.

A. Image Encoder–Decoder

The image encoder–decoder module [Fig. 4(a)] is crucial for extracting multiscale features from the input image, forming the foundation for subsequent tasks.

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1) Image Encoder: The backbone is a base Swin Transformer [76] pretrained on ImageNet [77]. The Swin Transformer excels in capturing multiscale features ($\mathcal{F} = \{\mathcal{F}_1, \mathcal{F}_2, \mathcal{F}_3, \mathcal{F}_4\}$) from the input image $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$, where $\mathcal{F}_i \in \mathbb{R}^{(H/2^{i+1})\times(W/2^{i+1})\times C_i}$.

2) Pixel Image Decoder: This component combines the feature pyramid network [78] and deformable detection Transformer [59], [79]. It processes the input image to produce four feature maps, each with distinct sizes: $(H/4)\times(W/4)$, $(H/8)\times(W/8)$, $(H/16)\times(W/16)$, and $(H/32)\times(W/32)$. These feature maps are collectively represented as $F = \{F_1, F_2, F_3, F_4\}$. To ensure compatibility with the original image dimensions, the pixel decoder upsamples these feature maps, resulting in output feature maps denoted as $F' = \{F'_1, F'_2, F'_3, F'_4\}$ and a mask feature $F_{\text{mask}} \in \mathbb{R}^{(H/32)\times(W/32)\times C_m}$.

B. Query Transformer Decoder

The query Transformer decoder, a crucial module designed for extracting building instances, processes pixel-level features to distill instance-specific attributes, including category, building boundaries, height, and offset vectors. It achieves this through a sophisticated architecture comprised of four cascaded cross-self-attention (CSA) modules. Each CSA module is equipped with masked cross-attention, self-attention, and feed-forward networks (FFNs), allowing for a nuanced interaction of instance-level features. Fig. 5 offers a comprehensive depiction of the CSA module's architecture, illustrating how it facilitates the extraction and integration of key attributes from the pixel-level inputs to instance-level outputs.

The computation process of this module is as follows.

1) *Initialization:* Generating learnable queries $\mathcal{Q}^{\text{init}} \in \mathbb{R}^{N_Q\times E_Q}$ and query position embeddings $e^q_{\text{pos}} \in \mathbb{R}^{N_Q\times E_Q}$. Here, N_Q represents the number of queries and also the number of target objects that the model can simultaneously detect, set at $N_Q = 200$. E_Q represents the dimensionality of the features used to characterize an instance, with $E_Q = 256$. Given that the self-attention mechanism in Transformers does not inherently process the order of elements within a sequence-calculating relationships among elements without knowledge of their arrangement—the creation of query position embeddings e^q_{pos} matching the dimensions of $\mathcal{Q}^{\text{init}}$ is crucial. These embeddings are designed to provide the sequence order of the N_Q instances and the order of the E_Q dimensional features within each instance.
2) *Masked Cross-Attention:* Calculating masked cross-attention between the queries and pixel-level features from the pixel decoder while masking out regions with low potential for containing targets. This allows each query to focus on potentially relevant local features. For detailed information on the masked cross-attention mechanism, please refer to the Appendix.
3) *Self-Attention:* Each query $\mathcal{Q}uery_i$ represents a potential target in a distinct local area. Calculating self-attention among these queries helps in capturing more semantically meaningful global features.
4) *Feature Generation:* An FFN is then employed to produce instance-specific features.

C. Multitask Decoder

The output from the query Transformer decoder, enriched with both local and global contextual understanding, generates refined queries. These queries, upon processing through distinct multilayer perceptrons (MLPs), yield various embeddings: mask embeddings $\mathcal{A}_M \in \mathbb{R}^{N_Q\times d_M}$, predicted logits $\mathcal{A}_C \in \mathbb{R}^{N_Q\times 2}$ for classification, offset vector embeddings $\mathcal{A}_O \in \mathbb{R}^{N_Q\times 2}$, height embeddings $\mathcal{A}_H \in \mathbb{R}^{N_Q\times d_M}$, and bin embeddings $\mathcal{A}_B \in \mathbb{R}^{N_Q\times d_M}$, with $d_M = 256$. This multifaceted output encapsulates key instance-specific attributes such as category, boundary delineation, offset vectors, and building heights.

Given the distinctive challenge of delineating building boundaries—a task that uniquely stands out among instance-based features due to its reliance on precise pixel-level information—the process begins with the transformation of initial pixel-level features F'_1 using an MLP to produce pixel embeddings $F_{\text{ebed}} \in \mathbb{R}^{E_Q\times(H/4)\times(W/4)}$. These embeddings match the dimensionality of mask embeddings $\mathcal{A}_M$, facilitating a seamless integration of low-level visual features with high-level instance attributes. To compute the predicted mask $\mathcal{M}_{\text{pred}} \in \mathbb{R}^{N_Q\times(H/4)\times(W/4)}$, a crucial step involves the matrix multiplication between $\mathcal{A}_M$ and F_{ebed}. This operation effectively merges the instance-level mask embeddings with the pixel embeddings, resulting in a spatially coherent representation that captures the segmented building instances within the image. Through this process, the model successfully translates the abstract, learned features into actionable, interpretable instance segmentation outputs

$$F_{\text{ebed}} = \text{MLP}\left(F'_1\right) \tag{3}$$

$$\mathcal{M}\text{pred} = \mathcal{A}_M \times F_{\text{ebed}}. \tag{4}$$

D. Height Decoder

In BDH-Net, the height decoder plays a pivotal role in extracting critical features from multimodal attributes of building instances and accurately predicting building heights. To enhance the precision of feature extraction, capture essential characteristics, and increase overall robustness, this module incorporates strategies like intrainstance attention, interinstance attention, and adaptive bin regression to enhance feature extraction and height prediction accuracy.

1) Location and Scale Attributions: Guided by Tobler's first law of geography [80], which posits that "everything is related to everything else, but near things are more related than distant things," it becomes clear that spatial relationships between objects are heavily influenced by their proximity. However, in the context of analyzing building instances within images, it is vital to extend this concept from the distance between two objects within the image to include the scale of the image as a factor influencing spatial relationships.

To address this, this research introduces the location attribute of instances and the scale attribute of images. The location attribute is represented by the geometric center coordinates $\mathcal{A}\text{coord} \in \mathbb{R}^{N_Q\times 2}$ of instance boundary predictions $\mathcal{M}_{\text{pred}}$. Concerning the scale attribute, with an initial image spatial resolution of $r_0^{\text{spatial}} = 0.54$ m and input image size of

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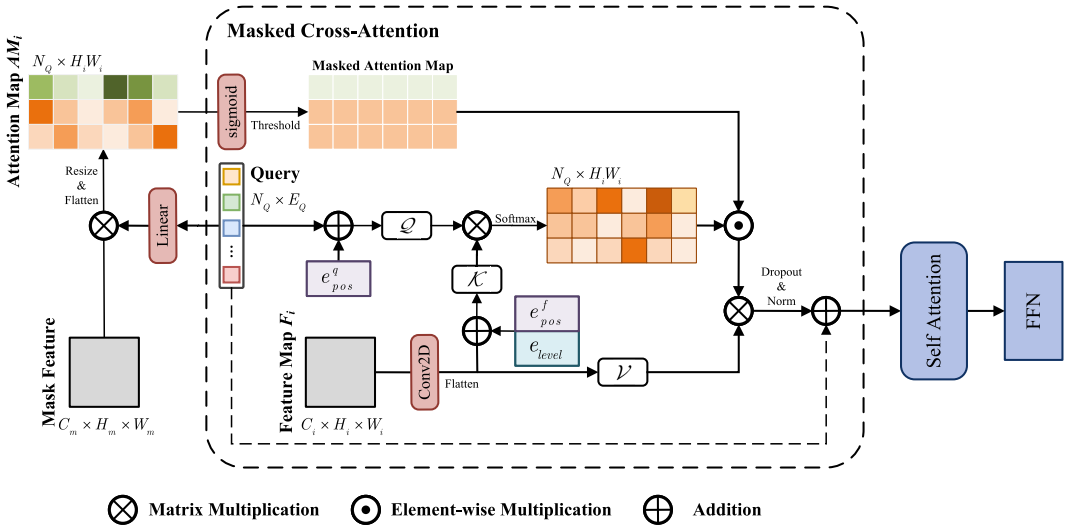


Fig. 5. Specific architecture of the proposed CSA in query Transformer decoder. This diagram simplifies the representation of various heads within the attention mechanism.

1024×1024, the scale is set to $\mathcal{S}_0 = 1.0$. This allows us to compute the scale for the input image as follows:

$$\mathcal{S} = \frac{\mathcal{S}_0 \times r_0^{\text{spatial}}}{\mathcal{R} \times r^{\text{spatial}}} \tag{5}$$

where $\mathcal{R} = W/W_0$, in which W represents the actual width of the image and $W_0 = 1024$, and r^{spatial} represents the spatial resolution of the input image.

2) Intrainstance Attention: The initial step involves identifying essential features within building instances. The multimodal instance attributions generated by the multitask decoder and query Transformer decoder are rich in information but may contain redundant data. To streamline this information and focus on what is truly relevant, we employ intrainstance attention on these multimodal instance attributes.

The input features encompass height embeddings $\mathcal{A}_H \in \mathbb{R}^{N_Q \times d_H}$, category logits $\mathcal{A}_C \in \mathbb{R}^{N_Q \times 2}$, mask embeddings $\mathcal{A}_M \in \mathbb{R}^{N_Q \times d_M}$, offset vector embeddings $\mathcal{A}_O \in \mathbb{R}^{N_Q \times 2}$, instance center coordinates $\mathcal{A}_{\text{coord}} \in \mathbb{R}^{N_Q \times 2}$, and a scale ratio $\mathcal{S} \in \mathbb{R}^{N_Q}$. These diverse components collectively form $\mathcal{A}_\mathcal{F}$

$$\mathcal{A}_\mathcal{F} = \text{Concat}(\mathcal{A}_H, \mathcal{A}_C, \mathcal{A}_M, \mathcal{A}_O, \mathcal{A}_{\text{coord}}, \mathcal{S}). \tag{6}$$

To recalibrate and refine these features, the intra-attention module processes $\mathcal{A}_\mathcal{F}$ as follows. First, the features undergo normalization and are then passed through an MLP to produce an intermediary feature, denoted as $\mathcal{A}'_\mathcal{F}$

$$\mathcal{A}'_\mathcal{F} = \text{Norm}(\text{MLP}(\mathcal{A}_\mathcal{F})). \tag{7}$$

Subsequently, these intermediary features are subject to a multihead attention mechanism, which recalibrates them based on the attention across different feature heads, yielding the recalibrated feature $\mathcal{A}_{\text{intra}} \in \mathbb{R}^{N_Q \times 256}$

$$\mathcal{A}_{\text{intra}} = \text{MultiHeadAttention}\big(\mathcal{A}'_\mathcal{F}\big). \tag{8}$$

Through this intrainstance attention mechanism, the model effectively focuses on the most significant features within each building instance, optimizing the representation of instance-specific attributes for interinstance attention.

3) Interinstance Attention Graph: Modeling the relationships between building instances is crucial for monocular height estimation, and it becomes even more critical for off-nadir monocular height estimation. This is due to the presence of occlusions caused by foreground buildings or shadows, which can prevent the accurate extraction of some buildings' offset vector information. In such scenarios, leveraging the principle that buildings within the same district or in close proximity are likely to have similar heights can significantly aid in estimating the height of obscured buildings.

In the context of graph neural networks [81], [82], the capability to align with the inherent topological structure of data plays a vital role. It not only allows for the preservation of localized structural information but also facilitates the effective learning of transitive relationships [83]. To address this, we introduce a novel module termed the IAG. As depicted in Fig. 6, IAG is conceived as an interinstance-level graph neural network tailored to encapsulate both spatial and attribute features, thereby augmenting the accuracy of instance segmentation and height estimation tasks. It achieves this by forming connections between all individual building instances viewed from an off-nadir perspective, enabling a comprehensive understanding of the complex interrelations and spatial dynamics among buildings. This approach not only enhances our model's ability to deal with occlusions effectively but also leverages the interconnected nature of urban buildings to improve the height estimation accuracy.

In formal terms, IAG can be represented as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of nodes and $\mathcal{E}$ represents the set of edges. Each node $v_i \in \mathcal{V}$ corresponds to a building instance, encapsulating multimodal aggregated features, including instance mask, label, location, roof-to-footprint offset, and scale. Each edge $e_{ij} \in \mathcal{E}$ signifies the relationship between two building instances, v_i and v_j, and can be represented as

$$e_{ij} = \frac{\exp\big(R_{i,j}\big)}{\sum \exp\big(R_{i,j}\big)} \tag{9}$$

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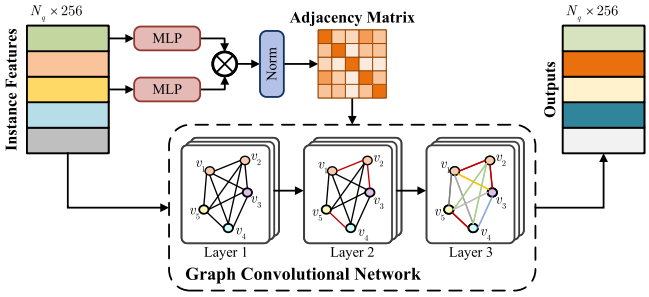


Fig. 6. Specific architecture of the proposed IAG.

$$R_{i,j} = f_{r1}(v_i) \cdot f_{r2}(v_i)^T \tag{10}$$

where f_{r1} and f_{r2} are two learnable MLPs and $R_{i,j}$ is the relation score between v_i and v_j.

Then, the graph neural network synthesizes the spatial and attribute relationships between nodes via a message-passing paradigm that propagates information along the edges. Ultimately, the output of IAG is processed to achieve feature $\mathcal{A}_{\text{inter}} \in \mathbb{R}^{N_Q \times 256}$ recalibration between instances. Through IAG, our model leverages the power of graph neural networks to embody a more sophisticated understanding of the spatial and relational context surrounding each building instance.

4) Adaptive Bin Height Regressor: The adaptive bin height regressor introduces an alternative approach to height estimation by conceptualizing the task as a classification problem, aiming to enhance the stability and accuracy of height predictions. This method contrasts with traditional fixed-bin methods, such as the deep ordinal regression network (DORN) [69], which employs a static number of bins of predetermined widths for depth partitioning. Instead, the adaptive bins method, inspired by [70], dynamically adjusts the binning strategy according to the specific features of the input scene. This approach combines the strengths of classification and depth map regression, predicting the final depth value as a linear combination of bin centers, thus promising greater precision and robustness in depth estimation. This concept has shown potential in the field of remote sensing for monocular height extraction [84].

In our adaptation of the method, originally designed for pixel-level prediction tasks with input feature $\mathcal{F} \in \mathbb{R}^{C \times H \times W}$, we refined it for instance-level height extraction and the input feature $\mathcal{A}_{\text{inter}} \in \mathbb{R}^{N_Q \times 256}$. Notably, we exclude the use of the softmax function on height features within the hybrid regression framework, a modification driven by its previously observed adverse effects on instance-level accuracy.

First, the adaptive bin embeddings $\mathcal{A}_B \in \mathbb{R}^{N_Q \times N_b}$ are generated by the query Transformer and the bin regressor, which is an MLP. They are subsequently processed through a LeakyReLU activation function and normalization to yield the final bins $\mathcal{B} \in \mathbb{R}^{N_q \times N_b}$

$$\mathcal{B}_i = \frac{\mathcal{A}_B + \epsilon}{\sum_{i=1}^{N_b} (\mathcal{A}_B + \epsilon)} \tag{11}$$

where ϵ is a small constant to avoid division by zero.

Next, the height for each bin center is computed using the following equation:

$$c(\mathcal{B}_i) = h_{\min} + (h_{\max} - h_{\min}) \cdot \left(\mathcal{B}_i/2 + \sum_{j=1}^{i-1} \mathcal{B}_j \right). \tag{12}$$

Ultimately, the predicted height $\mathcal{H} \in \mathbb{R}^{N_q}$ is obtained by multiplying the instance's height features with the corresponding bin center height information and then summing the results, as expressed in the following formula:

$$\mathcal{H}_i = \sum_{i=1}^{N_b} c(\mathcal{B}_i) \times \mathcal{A}_{\text{inter}}. \tag{13}$$

This refined approach, by accurately aligning the binning strategy with the unique attributes of each instance, significantly improves the model's capacity to estimate building heights with enhanced accuracy and adaptability.

E. Loss Functions

In the BDH-Net framework, we have developed a set of specialized loss functions to effectively address the challenges posed by building instance segmentation, offset vector estimation, and height estimation. These loss functions are designed to promote accuracy and alignment with ground truth data.

For the fundamental task of categorizing building instances, we employ binary cross-entropy (BCE) loss ($\mathcal{L}_{\text{bce}}$). This loss function quantifies the dissimilarity between predicted and ground truth binary labels, assessing how well our model classifies building instances.

The BCE loss is defined as

$$\mathcal{L}_{\text{bce}} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \cdot \log(p(y_i)) + (1 - y_i) \cdot \log(1 - p(y_i)) \right] \tag{14}$$

where N represents the total number of instance samples, y_i is the ground truth label for the ith sample, and $p(y_i)$ is the predicted probability of the ith sample belonging to the positive class.

To ensure accurate segmentation of building instances, we employ two complementary loss functions.

1) Sigmoid cross-entropy loss ($\mathcal{L}_{\text{mask}}$) encourages precise pixel-level predictions within the instance masks.
2) Dice loss ($\mathcal{L}_{\text{dice}}$) fosters alignment between predicted and ground truth instance masks by computing the Dice coefficient, which measures the overlap between the two masks.

The Dice loss is defined as

$$\mathcal{L}_{\text{dice}} = 1 - \frac{2 \sum_{i=1}^{N} p_i \cdot g_i}{\sum_{i=1}^{N} p_i^2 + \sum_{i=1}^{N} g_i^2}. \tag{15}$$

Precise localization of building instances is achieved through the prediction of offset vectors, and accurate estimation of building heights is crucial. Therefore, we employ both $L1$ and $L2$ loss functions to assess disparities between predicted and true values.

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1) $L1$ loss emphasizes the absolute differences between predicted and true offset vectors

$$\mathcal{L}_{l1} = \frac{1}{N}\sum_{i=1}^{N}\left|v_i - \hat{v}_i\right| \tag{16}$$

where v_i represents the predicted offset vector or height values and $\hat{v}_i$ represents the reference offset vector or height values.

2) $L2$ loss focuses on the squared deviations between predicted and true offset vectors

$$\mathcal{L}_{l2} = \frac{1}{N}\sum_{i=1}^{N}\left(v_i - \hat{v}_i\right)^2. \tag{17}$$

Our BDH-Net model effectively handles the complexities of building instance segmentation, localization, and height estimation by combining these tailored loss functions. The total loss, denoted as $\mathcal{L}_{\text{total}}$, is expressed as a linear combination of different loss components

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{bce}} + \beta \cdot \mathcal{L}_{\text{mask}} + \gamma \cdot \mathcal{L}_{\text{dice}} + \delta \cdot \mathcal{L}_{\text{offset}} + \theta \cdot \mathcal{L}_{\text{height}} \tag{18}$$

where α, β, γ, δ, and θ are weights dynamically adjusted during different stages of model training to optimize the performance. In the initial training stage, the model focuses on category classification and mask instance segmentation, setting the weights as follows: $\alpha = 5.0$, $\beta = 200.0$, $\gamma = 5.0$, $\delta = 0.001$, and $\theta = 0.001$. Once the model achieves reliable recognition of building instances, it shifts its focus toward optimizing offset vector and height loss functions, adjusting the weights as follows: $\alpha = 5.0$, $\beta = 200.0$, $\gamma = 1.0$, $\delta = 0.1$, and $\theta = 0.2$. This dynamic weight adjustment enables the model to adapt and improve its performance over time, effectively addressing the complex tasks of building instance segmentation, localization, and height estimation.

V. Experiment Settings

A. Evaluation Metrics

This study conducts a comprehensive evaluation of building segmentation and height estimation, focusing on both boundary quality and height prediction accuracy.

For assessing building instance segmentation, we utilize average precision (AP) in conjunction with AP_{50} and AP_{75} as instance-level metrics. These metrics gauge the model's ability to detect and accurately delineate individual building instances. AP provides a summary of segmentation performance across varying confidence thresholds by calculating the area under the precision–recall curve. AP_{50} and AP_{75} represent the AP values at critical IoU thresholds of 0.5 and 0.75, respectively. Following the standard COCO evaluation protocol, $AP(\tau)$ is computed at multiple IoU thresholds, denoted as $\mathcal{T} = \{0.50, 0.55, \ldots, 0.95\}$, and then averaged to obtain the AP:

$$\text{AP}(\tau) = \int_0^1 p(r;\tau)dr \tag{19}$$

$$\text{AP} = \frac{1}{|\mathcal{T}|}\sum_{\tau\in\mathcal{T}}\text{AP}(\tau) \tag{20}$$

where $p(r;\tau)$ represents the precision at recall r using an IoU threshold $> \tau$ to determine true and false positives.

For quantitatively evaluating building height prediction, a combination of error and accuracy metrics is employed. The mean average error (MAE) and root-mean-square error (RMSE) serve as error metrics, with lower values indicating superior performance. The threshold metric δ assesses height accuracy by quantifying the percentage of predictions falling within specified tolerance levels of the ground truth. These metrics are defined as follows:

$$\text{MAE} = \frac{1}{N}\sum_{h\in N}\left|y_i - \hat{y}_i\right| \tag{21}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{h\in N}\left(y_i - \hat{y}_i\right)^2} \tag{22}$$

$$\text{acc} = \frac{N_h}{N} \quad \text{s.t.}: \max\left(\frac{\hat{h}}{h}, \frac{h}{\hat{h}}\right) = \delta_i \le 1.25^i, \ i = 1, 2, 3 \tag{23}$$

where N represents the total evaluated instances and $\hat{h}$ and h denote the predicted and ground truth heights, respectively.

B. Experimental Settings

For all experiments, we establish a virtual Anaconda environment with Python 3.7 and PyTorch 1.8.2 as the standard. The specific graph computation platform includes CUDA 11.1 and CUDNN 8.0.4, executed on three NVIDIA RTX 3090 GPUs. The batch size is set to 1, and the learning rate is initialized at 1×10^{-5}. The model undergoes training for 80 000 iterations, with the learning rate reduced by a factor of 0.1 at the 8000th and 64 000th epochs. The optimization process employs the AdamW optimizer with a weight decay set to 0.05. Validation is performed on the validation set of the BDH dataset.

C. Comparison Methods

Given the limited existing research on building height estimation from off-nadir satellite sensor imagery by instance segmentation, we devised comparative methods denoted as $\mathcal{M}_l$ and $\mathcal{M}_p$ based on a crucial finding in building height estimation: the linear correlation between building height and the magnitude of the offset vector [22]. These methods provide baseline references for our proposed approach.

The linear regression method $\mathcal{M}_l$ is

$$\mathcal{M}_l(\mathcal{O}) = 0.542\mathcal{O} + 8.409 \tag{24}$$

where $\mathcal{O}$ is the offset vector.

The polynomial regression method $\mathcal{M}_p$ is

$$\mathcal{M}_p(\mathcal{O}) = -0.001\mathcal{O}^2 + 0.690\mathcal{O} + 5.735 \tag{25}$$

where $\mathcal{O}$ is the offset vector.

Furthermore, to further benchmark our model's performance, we incorporated the height estimation Baseline[2] from Track 2 of the 2023 IEEE GRSS Data Fusion Contest, titled "Multitask Learning of Joint Building Extraction and Height

[2]https://github.com/AICyberTeam/DFC2023-baseline/tree/main/track2

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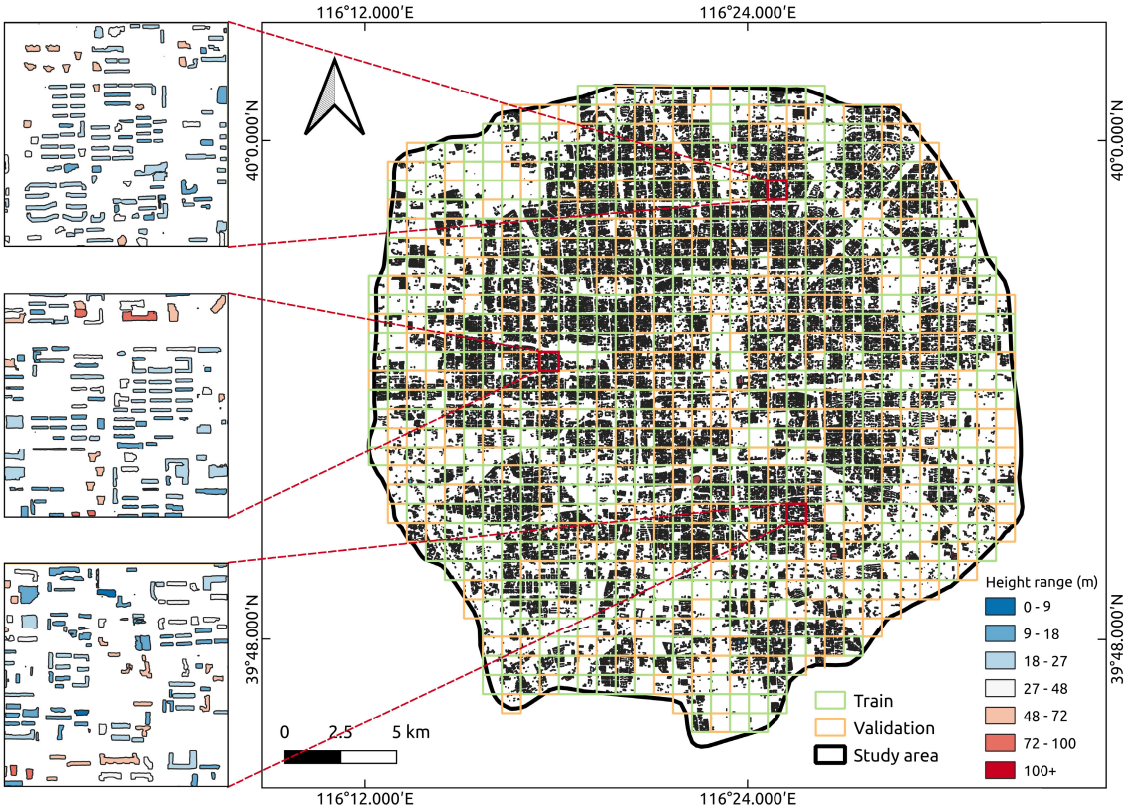


Fig. 7. Comprehensive visualization of the extracted 3-D building information across the study area.

Estimation," to construct comparative methods alongside Mask R-CNN [85], Cascade Mask R-CNN [86], TensorMask [87], MaskFormer [60], and Refactored MaskFormer [85]. While these instance segmentation networks were employed to extract instance-level boundary information of buildings, they were not used directly for height estimation. Also, we add the method LOFT-FOA [88], which also uses the offset vector to the comparison.

VI. Results

This section delineates a comprehensive evaluation of BDH-Net's performance across two key subtasks: building height estimation and building boundary recognition. We start with an evaluation of building boundary recognition, followed by an analysis of the model's proficiency in building height estimation. An overview of the 3-D building information is presented in Fig. 7. For a detailed display of 3-D building models specific to the Haidian District, please see Fig. 8. Detailed representations for other districts in the 5th Ring Road region of Beijing are provided in Figs. 9–12.

A. Building Boundary Instance Segmentation Results

The performance of BDH-Net in precisely demarcating building boundaries is a focus of this study, showing the model's capability to recognize the intricate contours of urban landscapes. Table II presents BDH-Net achieving an AP of 34.5, with high AP_{50} and AP_{75} scores of 56.4 and 38.9, respectively. Table II highlight BDH-Net's notable accuracy in outlining buildings, positioning it as a strong contender among established segmentation techniques like Mask R-CNN, Cascade Mask R-CNN, and TensorMask.

TABLE II
Comparative Analysis of Instance Segmentation Metrics

Method	AP	AP_{50}	AP_{75}
Mask R-CNN	24.2	42.6	26.1
Cascade Mask R-CNN	34.0	55.8	38.5
TensorMask	19.9	42.5	16.0
MaskFormer	32.8	53.8	36.6
Refactored MaskFormer	32.0	54.8	35.1
LOFT-FOA	33.0	54.0	37.0
BDH-Net	34.5	56.4	38.9

BDH-Net slightly outshines Cascade Mask R-CNN in AP and AP_{50}, demonstrating its broad efficacy in instance segmentation. In particular, BDH-Net's AP_{75} score reveals its high precision in achieving highly precise boundary delineations at more stringent IoU thresholds. Outperforming Mask R-CNN and TensorMask, this achievement emphasizes BDH-Net's ability to segment urban edifices with remarkable precision crucial for complex urban modeling and detailed analytical tasks.

The lesser performance observed in Mask R-CNN and TensorMask may stem from the challenges inherent in multitask learning, where balancing instance segmentation with height estimation presents considerable difficulties. These models struggle due to the minimal interaction between boundary and height features during intermediate stages of feature extraction. The absence of such feature interplay, coupled

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Fig. 8. 3-D visual presentation of Haidian District.



Fig. 9. 3-D visual presentation of Xicheng District.



Fig. 10. 3-D visual presentation of Dongcheng District.

with the lack of supervision from intermediary features like offset vectors, compounds these difficulties. These insights highlight the importance of incorporating multifeature interactions at the instance level and leveraging offset vectors as surrogate features, which are key to enhancing model performance, especially in tasks that require meticulous attention to both the dimensionality and spatial segmentation of urban forms. Conversely, the cascading architecture of Cascade Mask

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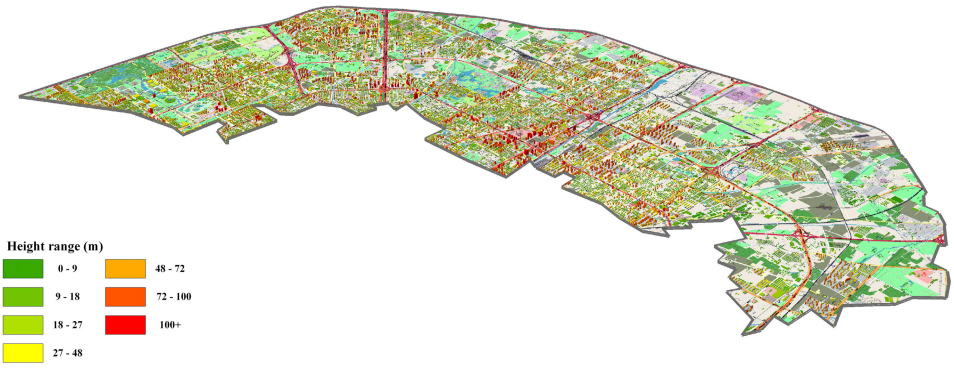


Fig. 11. 3-D visual presentation of Chaoyang District.



Fig. 12. 3-D visual presentation of the rest districts.

R-CNN inherently facilitates a more effective gradient flow, allowing earlier layers to receive more precise gradient signals. This characteristic inherently enhances performance in tasks that necessitate sequential refinement, such as object detection and segmentation, by providing a structured approach to progressively refine predictions at each stage. Such an approach proves advantageous in multitask settings, like those involving both segmentation and height estimation, where the complexity of tasks demands nuanced handling.

For a more tangible illustration of BDH-Net's boundary delineation capabilities, Fig. 13 visually presents the detection outcomes, with activation maps showing the model's focus and identification of structural details, further validating BDH-Net's proficiency in urban structure segmentation.

Fig. 14 offers a comparative visual presentation of 3-D building information extraction across three distinct urban landscapes, detailing the efficacy of different models in this domain. The cases include an area of irregularly distributed high-rise buildings (Case 1), a zone of regularly patterned low-rise buildings (Case 2), and an area of irregularly spaced low-rise structures at the base (Case 3). In the critical task of extracting building roof boundaries, BDH-Net demonstrates notable strengths in handling the complexity of urban scenes, with significantly fewer omissions in detection compared to other methods.

B. Building Height Results

The predictive performance regarding building heights is of particular interest in our study. From an intuitive visual perspective, Fig. 14 shows the results of 3-D building information extraction, clearly indicating that BDH-Net yields more accurate height predictions. This is especially apparent for high-rise buildings, where BDH-Net's estimations are distinctly more precise when compared to other deep learning-based methods, including Cascade Mask R-CNN and Mask R-CNN.

To further quantify this precision in building height predictions, an analytical classification was conducted in accordance with Chinese Architectural Design Fire Protection Code GB50016-2014 (2018 edition). This standard divides buildings into low-rise (0–27 m), high-rise (27–100 m), and super-high-rise (more than 100 m) categories. A detailed performance evaluation of the model's height estimations within these defined categories is methodically tabulated in Table III. The results are analyzed as follows.

1) Low-Rise Buildings: BDH-Net significantly outperforms the regression methods ($\mathcal{M}_l$ and $\mathcal{M}_p$) in estimating the heights of low-rise buildings, with an MAE of 3.61 m and an RMSE of 5.24 m. Its performance notably exceeds that of traditional deep learning approaches by at least 0.6 m in MAE. Notably, MaskFormer (an MAE of 3.95 m), Refactored MaskFormer (an MAE of 3.71 m), and LOFT-FOA (an MAE

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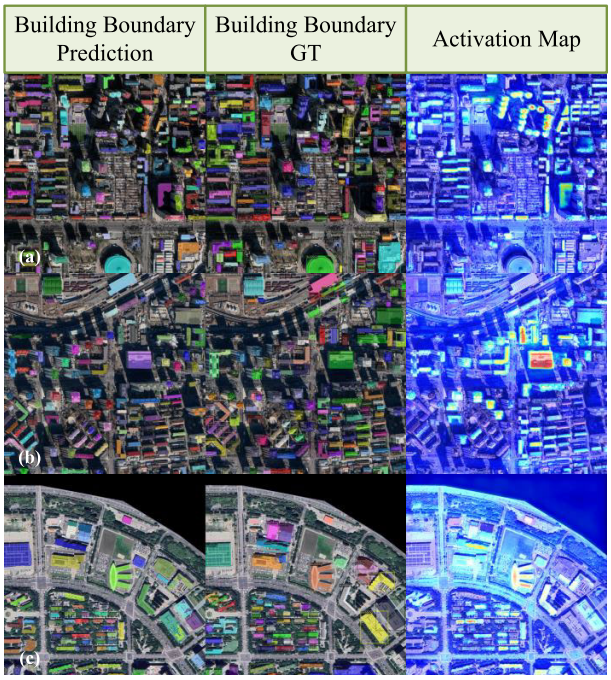


Fig. 13. Visual presentation of predicted versus actual building roof boundaries and corresponding activation maps across diverse regions, illustrating the model's focus on structural details. (a)–(c) Model's performance in predicting building boundaries and revealing the activation maps for building instances within pixel embeddings.

TABLE III

COMPARATIVE ANALYSIS OF HEIGHT ESTIMATION METRICS ACROSS DIFFERENT METHODS

Method	Height Range (m)	MAE (m)	RMSE (m)	δ_1	δ_2	δ_3
M_l	0-27	5.74	8.26	0.468	0.712	0.808
M_p		5.76	8.28	0.444	0.701	0.808
Mask R-CNN		4.36	6.33	0.733	0.963	0.993
Cascade Mask R-CNN		4.30	6.42	0.567	0.822	0.913
TensorMask		4.48	6.71	0.571	0.824	0.910
MaskFormer		3.95	5.87	0.628	0.872	0.936
Refactored MaskFormer		3.71	5.45	0.646	0.881	0.942
LOFT-FOA		3.74	5.13	0.598	0.869	0.939
BDH-Net		3.61	5.24	0.659	0.883	0.944
M_l	27-100	15.13	19.53	0.343	0.669	0.837
M_p		14.34	18.94	0.390	0.707	0.847
Mask R-CNN		10.81	14.42	0.618	0.884	0.958
Cascade Mask R-CNN		10.30	14.05	0.621	0.883	0.955
TensorMask		11.76	15.56	0.564	0.835	0.910
MaskFormer		9.96	13.04	0.630	0.898	0.973
Refactored MaskFormer		9.38	12.34	0.668	0.913	0.971
LOFT-FOA		8.84	12.03	0.700	0.910	0.969
BDH-Net		7.94	10.90	0.758	0.936	0.977
M_l	100+	33.60	51.32	0.467	0.717	0.751
M_p		30.82	41.77	0.540	0.749	0.810
Mask R-CNN		47.48	61.28	0.250	0.625	0.750
Cascade Mask R-CNN		48.08	64.09	0.314	0.712	0.843
TensorMask		47.27	58.96	0.214	0.643	0.727
MaskFormer		54.78	67.74	0.194	0.484	0.710
Refactored MaskFormer		43.60	57.60	0.333	0.666	0.727
LOFT-FOA		42.68	61.68	0.438	0.656	0.781
BDH-Net		34.79	49.10	0.364	0.771	0.881
M_l	Total	8.86	14.05	0.432	0.700	0.815
M_p		8.59	13.32	0.430	0.703	0.818
Mask R-CNN		6.13	9.61	0.584	0.846	0.925
Cascade Mask R-CNN		6.04	9.36	0.602	0.853	0.930
TensorMask		6.30	9.93	0.569	0.872	0.927
MaskFormer		5.83	9.06	0.628	0.879	0.947
Refactored MaskFormer		5.54	8.62	0.652	0.891	0.951
LOFT-FOA		5.38	8.33	0.628	0.881	0.948
BDH-Net		5.00	7.77	0.689	0.899	0.954

of 3.74 m) demonstrate good performance in low-rise building height estimation, yet still fall short of BDH-Net's precision. This superior accuracy highlights BDH-Net's ability to effectively interpret dense urban areas and complex ground-level interactions, crucial for applications in urban planning and development assessments.

2) High-Rise Buildings: BDH-Net's performance is particularly strong, with its MAE lead over the second-best method increasing from 15% in the low-rise category to

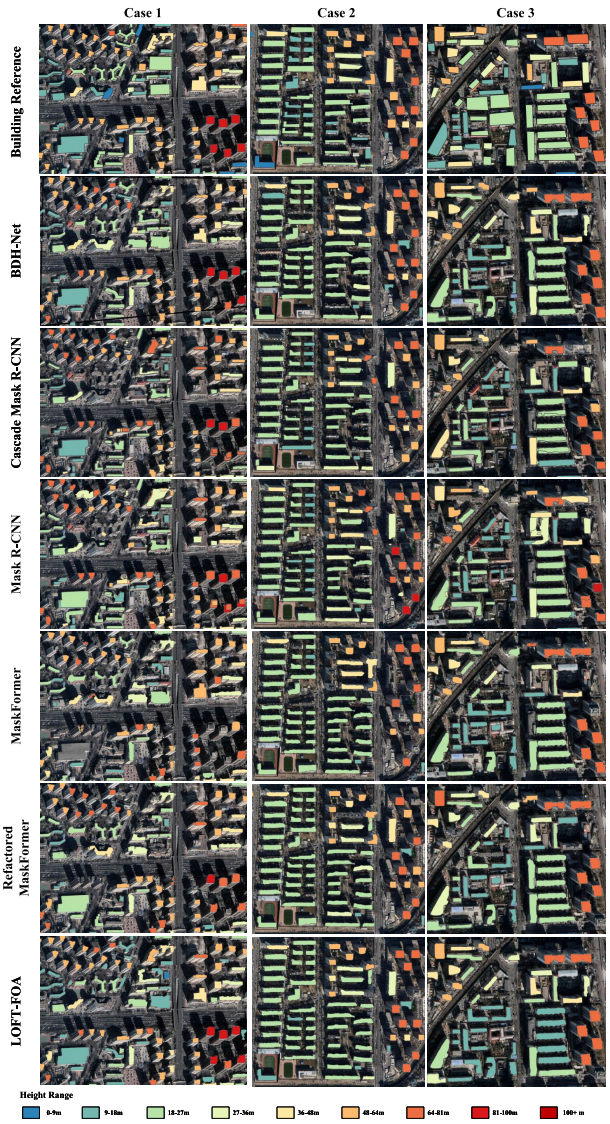


Fig. 14. Visual presentation of building 3-D information.

23%. The regression methods perform even worse in this category, with none achieving an MAE under 10 m. Deep learning methods also fall short of this benchmark. In the high-rise building category, MaskFormer (an MAE of 9.96 m), Refactored MaskFormer (an MAE of 9.38 m), and LOFT-FOA (an MAE of 8.84 m), while outperforming other baseline methods, still show a significant gap compared to BDH-Net (MAE of 7.94 m). BDH-Net's extended advantage in high-rise buildings is likely due to its sophisticated utilization of offset vectors and instance relationship modeling, enabling a more accurate interpretation of the vertical dimensions of urban structures.

3) Super-High-Rise Buildings: The accuracy diminishes for super-high-rise buildings, where BDH-Net records an MAE of 34.79 m and an RMSE of 49.10 m. In this category, MaskFormer performs worst with an MAE of 54.78 m, while Refactored MaskFormer (an MAE of 43.60 m) and LOFT-FOA (an MAE of 42.68 m), though better than other deep learning methods, still underperform compared to BDH-Net and regression methods. This decline underscores the challenges

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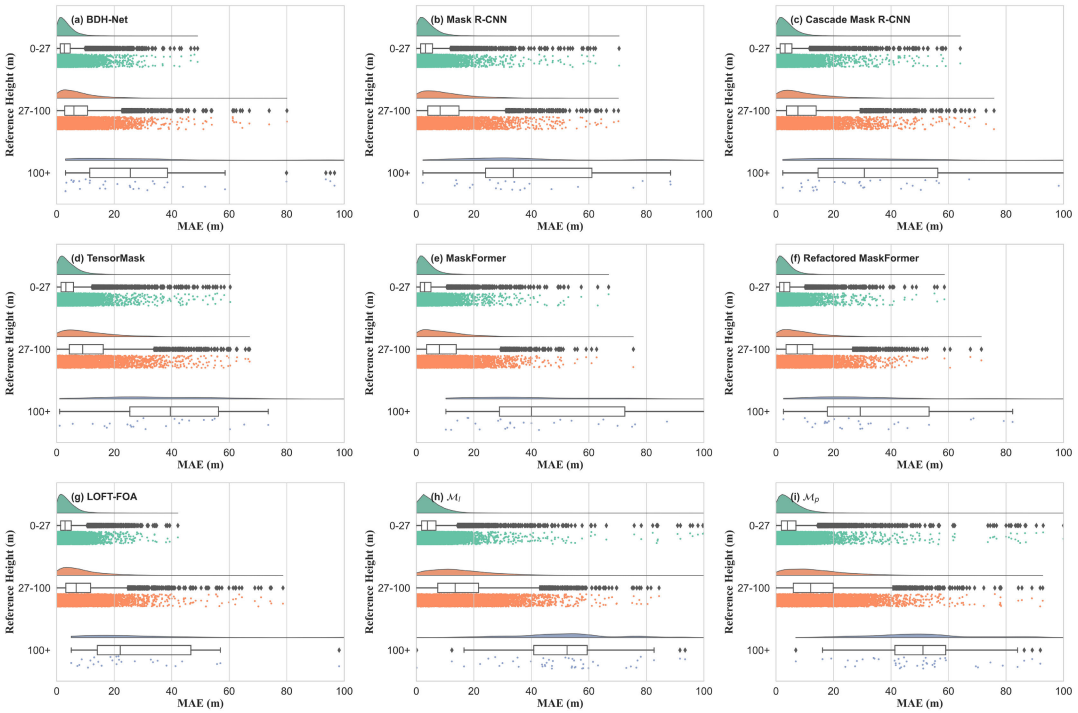


Fig. 15. MAE raincloud plot. (a) BDH-Net results. (b) Mask R-CNN results. (c) Cascade Mask R-CNN results. (d) TensorMask results. (e) Linear regression method $\mathcal{M}_l$ results. (f) Polynomial regression method $\mathcal{M}_p$ results. (g) LOFT-FOA. (h) $\mathcal{M}_l$. (i) $\mathcal{M}_p$.

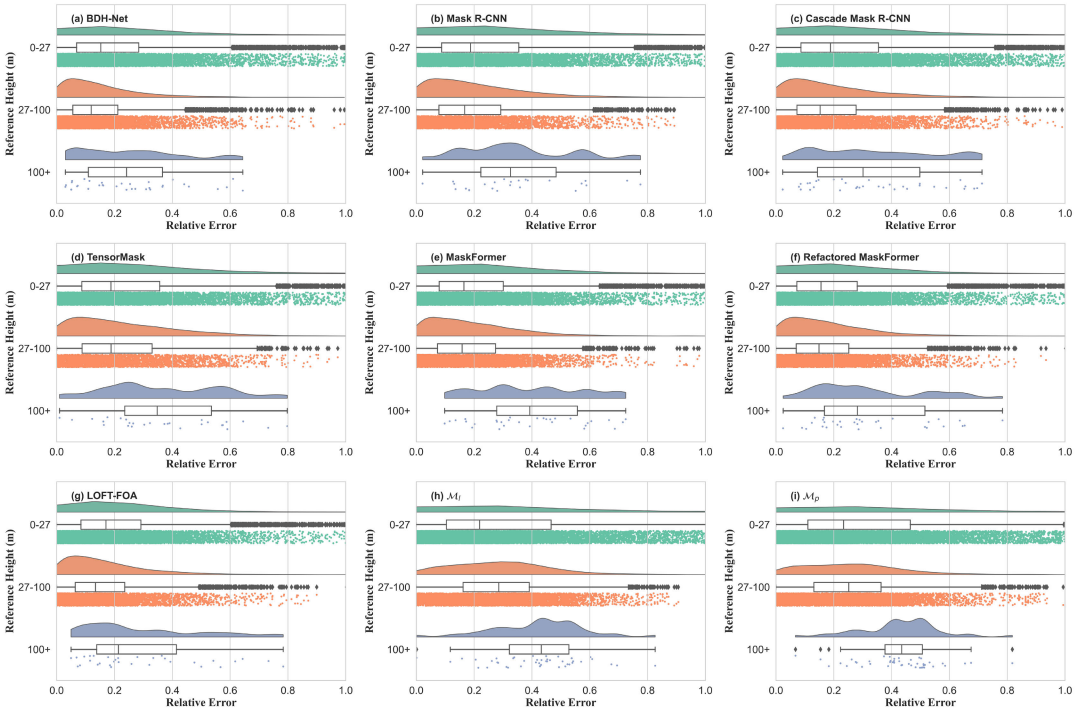


Fig. 16. Relative error raincloud plot. (a) BDH-Net results. (b) Mask R-CNN results. (c) Cascade Mask R-CNN results. (d) TensorMask results. (e) Linear regression method $\mathcal{M}_l$ results. (f) Polynomial regression method $\mathcal{M}_p$ results. (g) LOFT-FOA. (h) $\mathcal{M}_l$. (i) $\mathcal{M}_p$.

posed by the limited representation of such buildings in the training dataset, which impacts the model's ability to learn effectively from this particular segment. Additionally, the complexity of accurately capturing the architectural nuances of super-high-rise structures—such as their slender profiles or the presence of spires and antennas—contributes to the larger discrepancies in height predictions. These challenges highlight a need for enhanced training strategies and possibly more advanced model architectures to improve predictions for towering structures.

In summary, BDH-Net's outperformance across low- and high-rise categories demonstrates its effectiveness in urban

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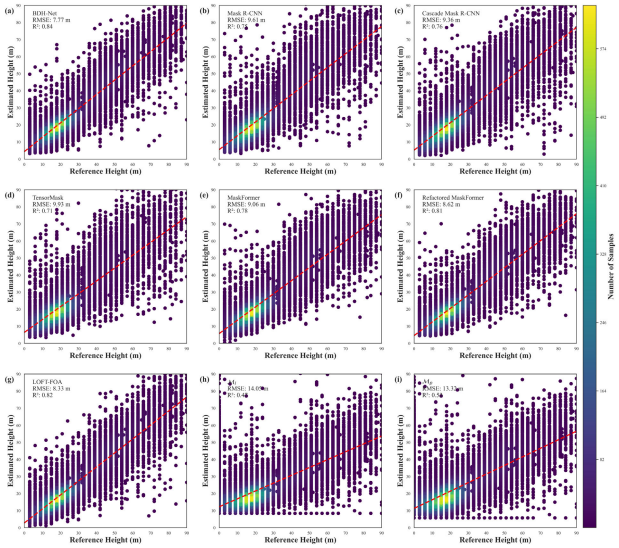


Fig. 17. Comparative density scatter plots of building height predictions obtained by (a) BDH-Net, (b) Mask R-CNN, (c) Cascade Mask R-CNN, (d) TensorMask, (e) MaskFormer, (f) Retrofitted MaskFormer, (g) LOFT-FOA, and (h) M_l, (m) M_p.

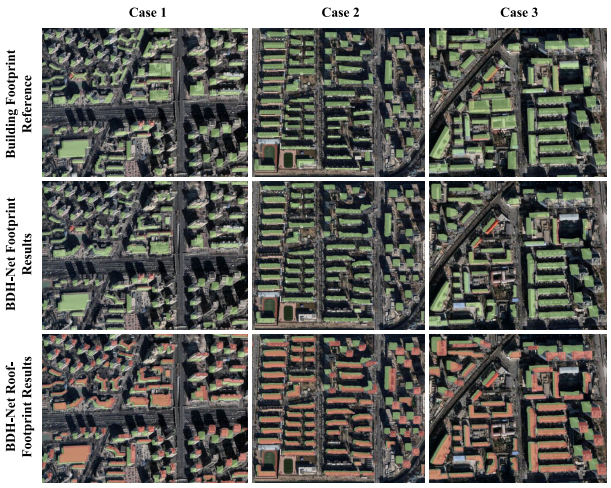


Fig. 18. Visual presentation of building footprint extraction by BDH-Net.

height estimation, benefiting from advanced feature extraction and spatial relationship analysis. However, the diminished accuracy in the super-high-rise category indicates areas for further development, emphasizing the need for dataset enrichment and model refinement.

Figs. 15 and 16 provide a more nuanced interpretation of height prediction results. As evidenced in Fig. 15, not only does BDH-Net have the smallest MAE, but its distribution is also tighter, which attests to the method's superior precision and consistency. Moreover, deep learning-based approaches generally outperform regression methods based on offset vectors, suggesting that neural networks' exceptional capabilities in feature extraction and representation significantly enhance the accuracy of building height extraction. Correspondingly, as shown in Fig. 16, the relative error in height predictions echoes similar conclusions to those drawn from Fig. 15.

The scatter plot in Fig. 17 corroborates these findings, highlighting the limitations of simple linear and polynomial models that tend to significantly underestimate building heights. This underestimation is reflected in their lower R^2 values, indicating weaker correlations between predicted and actual heights. Although polynomial regression can approximate certain nonlinear patterns, our results show that it struggles to generalize to buildings outside the training data distribution. This demonstrates that regression-based methods are insufficient for the building height estimation task. In contrast, BDH-Net, with its deep learning-based approach, consistently achieves higher R^2 values, capturing complex relationships with greater precision and exhibiting superior robustness and generalizability.

One of the challenges in extracting building heights from off-nadir imagery, combined with instance segmentation, lies in accurately restoring the building's footprint from its roof to determine its positional information. Fig. 18 shows a visual representation of building footprint extraction results by BDH-Net, where green indicates the building footprint and red marks the roof. The illustration reveals that the extraction of footprints is considerably accurate, reflecting BDH-Net's effectiveness in capturing the essential architectural details necessary for precise urban mapping.

C. Ablation Study

In this ablation study, the role and influence of distinct features in the BDH-Net model are dissected to determine their individual and collective contributions to height prediction performance. Table IV provides a detailed performance evaluation of BDH-Net across various configurations, illustrating the effectiveness of offset vectors, scale integration, IAG, and the adaptive bins (Adabins) regression strategy.

Findings indicate that the use of offset vectors as height indicators, coupled with scale sensitivity, greatly improves the model's height prediction accuracy. Both IAG and Adabins positively influence the height estimation capabilities of BDH-Net, as evidenced by their performance. Notably, as shown in Table V, the computational complexity of IAG and Adabins accounts for only a small proportion of the overall model, highlighting the efficiency of these modules. The conjunction of offset vectors and scale data emerges as particularly potent, substantially elevating the model precision in extracting building boundaries. This improvement suggests that the coordinated interaction between offset vectors and scale measurements considerably strengthens the model's overall performance, demonstrating a reciprocal enhancement between accurate height prediction and boundary detection while maintaining computational efficiency.

Table VI presents the performance comparison of different methods using ResNet-50 and ViT-S as backbones, evaluated in terms of MAE, RMSE, and threshold accuracies (δ_1–δ_3). The results demonstrate that BDH-Net consistently outperforms the baseline methods (Mask R-CNN, Cascade Mask R-CNN, and TensorMask) across all metrics. Specifically, BDH-Net achieves lower MAE and RMSE values while attaining higher threshold accuracies, particularly when using the ViT-S backbone, where its performance further improves. Additionally, it is observed that methods based on ViT-S generally outperform those using ResNet-50, indicating that the Transformer-based ViT-S backbone can provide more

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TABLE IV
PERFORMANCE EVALUATION OF BDH-Net FOR ABLATION STUDY

Method	Offset	Scale	IAG	Adabins	MAE (m)	RMSE (m)	δ_1	δ_2	δ_3	AP	AP_{50}	AP_{75}
BDH-Net	✗	✓	✓	✓	5.96	9.38	0.609	0.872	0.944	30.4	51.6	33.1
BDH-Net	✓	✗	✓	✓	5.87	9.19	0.628	0.879	0.947	30.5	51.9	33.4
BDH-Net	✓	✓	✗	✓	5.36	8.33	0.628	0.881	0.948	32.6	53.6	36.7
BDH-Net	✓	✓	✓	✗	5.23	8.28	0.650	0.885	0.949	33.0	53.9	37.2
BDH-Net	✓	✓	✓	✓	5.00	7.77	0.689	0.899	0.954	34.5	56.4	38.9

TABLE V
COMPUTATIONAL COMPLEXITY (IN FLOPS) OF EACH MODULE AND THEIR RELATIVE PERCENTAGES

Module	FLOPs (M)	Proportion
Image Encoder	48.8	69.3
Image Decoder	5.4	7.7
Query Transformer Decoder	14.3	20.3
Multi-task Decoder	0.4	0.6
Intra-instance Attention	0.3	0.4
IAG	0.8	1.1
Adabins	0.4	0.6
Total	70.4	100

effective feature representations for complex tasks such as building height estimation and segmentation. These findings highlight the effectiveness of BDH-Net's design and its ability to leverage advanced backbone architectures for improved performance and robustness.

Our investigation delves into the nuances of adaptive bins (Adabins) parameter configurations and their variants. The results, meticulously outlined in Table VII, contrast the Adabins implementation in this research with the prototype introduced in [70]. This comparison elucidates the effectiveness of the refinements and methodological adjustments we have implemented, transitioning from pixel-level to instance-level analyses.

Furthermore, the study probes into the effect of varying the number of bins on the accuracy of building height estimation. The comparative analysis is systematically presented in Table VIII, which reveals that changes in the bin count exhibit a relatively minor influence on the overall results. This finding led to the selection of a 256-bin configuration for our study, a choice underscored by its consistent provision of optimal performance across the evaluations conducted.

D. Inference Efficiency

The BDH-Net achieves an inference time of 0.168 s for a single image (0.85 km × 0.85 km) on an NVIDIA RTX 3090 GPU, which translates to an inference speed of approximately 0.233 s/km^2. This impressive inference speed paves the way for the potential expansion of this method to research endeavors encompassing the entirety of China (about 25.8 days).

E. Comparison With Other Urban Height Products

The field of building height extraction showcases a variety of methodologies as evidenced in studies in [49], [89], and [90]. Each method brings unique insights into resolution, accuracy, and practicality for urban mapping, a comparison of which is detailed in Fig. 19.

Our analysis reveals distinct limitations in the approaches of [49] and [89]. The approach in [49], in particular, tends to underestimate the heights of urban structures, often not exceeding 36 m. This pattern, as shown in Fig. 19, is especially incongruent in urban environments characterized by high-rise buildings, suggesting a gap in Wu et al.'s approach [49] for such contexts. The approach in [89], meanwhile, is hindered by its lower resolution outputs, limiting its capacity to depict the detailed urban landscape necessary for comprehensive planning.

Our research adopts a more refined approach to height extraction. Although we converted our high-resolution vector data (0.54 m) into a 1-m raster format for comparison purposes, the primary focus of our methodology is on the precision and detail it affords. This is clearly illustrated in Fig. 19, where our approach excels in identifying individual buildings and accurately capturing their heights, even in densely built urban areas. The meticulousness of our method allows for a more refined and accurate representation of urban landscapes, significantly improving upon the block-level analyses provided by Wu and He's studies. The high-resolution nature of our data is crucial for effective urban planning and development, as it enables a detailed understanding of complex urban patterns. Furthermore, our results, particularly regarding height patterns, trends, and range of values, align closely with the findings in [90], but with a finer resolution. This similarity not only affirms the precision of our methodology but also underscores its enhanced capacity for detailed spatial analysis in urban landscape mapping.

VII. DISCUSSION

A. Limitations

While the BDH-Net approach exhibits promising results in the extraction of 3-D building information, it occasionally encounters challenges as depicted in specific samples.

Challenge 1 (Shadow Occlusion): This limitation is inherent to the approach. It manifests when there is a notable tilt during satellite acquisitions, especially in densely built-up areas with high-rise structures. As illustrated in Fig. 20(a), the model adeptly extracts the building rooftops (denoted by the red box) with commendable boundary precision. However, extensive shadowing of the building's lower portion compromises the accuracy in estimating the offset vector's length and orientation, leading to notable height prediction discrepancies.

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TABLE VI
PERFORMANCE COMPARISON OF DIFFERENT METHODS WITH VARYING BACKBONES

Method	Backbone	MAE (m)	RMSE (m)	δ_1	δ_2	δ_3
Mask R-CNN	ResNet-50	6.13	9.61	0.584	0.846	0.925
Cascade Mask R-CNN		6.04	9.36	0.602	0.853	0.930
TensorMask		6.30	9.93	0.569	0.872	0.927
BDH-Net		5.46	8.35	0.653	0.891	0.941
Mask R-CNN	ViT-S	6.02	9.35	0.603	0.850	0.931
Cascade Mask R-CNN		5.76	8.86	0.631	0.873	0.936
TensorMask		6.15	9.66	0.582	0.877	0.929
BDH-Net		5.00	7.77	0.689	0.899	0.954

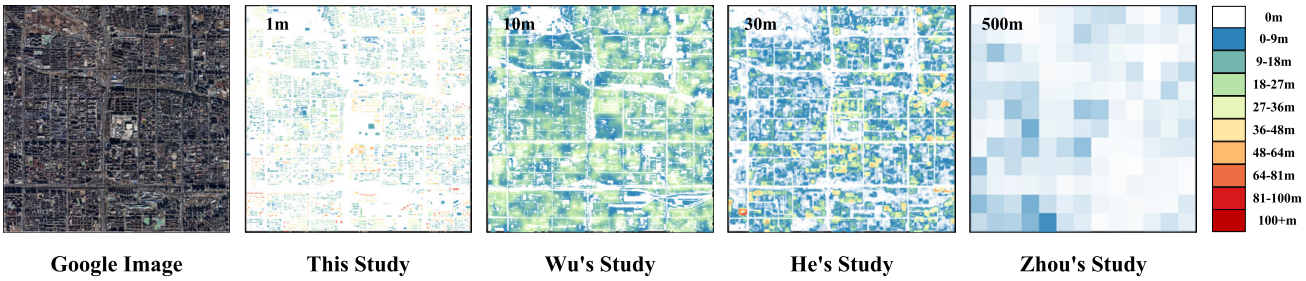


Fig. 19. Comparison with other multiscale building height products.

TABLE VII
PERFORMANCE COMPARISON BETWEEN DIFFERENT TYPES OF ADAPTIVE BINS

Bin Type	MAE (m)	RMSE (m)	AP	AP_{50}	AP_{75}
[70]	6.65	12.61	29.2	49.8	31.8
proposed	5.00	7.77	34.5	56.4	38.9

TABLE VIII
IMPACT OF VARYING NUMBER OF BINS ON HEIGHT ESTIMATION

Bin Channel	MAE (m)	RMSE (m)	δ_1	δ_2	δ_3
64	5.09	8.03	0.671	0.894	0.951
128	5.14	8.08	0.658	0.889	0.948
256	5.00	7.77	0.689	0.899	0.954
512	5.21	8.24	0.663	0.890	0.949



Fig. 20. Challenging cases. (a) Shadow occlusion. (b) Image stitching issues.

Challenge 2 (Image Stitching): When the research area coincides with the image stitching boundary of Google satellite imagery, variations in factors like off-nadir angles and solar elevation angle can induce pronounced differences in offset directions, as evident in Fig. 20(b). Inconsistencies in image radiance, color, and resolution further complicate the prediction process.

Challenge 3 (Super High-Rise Buildings): Table III indicates that both deep learning-based methods and regression approaches exhibit significant errors when estimating the heights of super high-rise buildings. This discrepancy primarily stems from the limited number of super high-rise buildings in the dataset, which hinders the model's ability to effectively learn their unique features. Additionally, the inherent complexity of super high-rise buildings, such as slender contours, spires, and the presence of antennas, further contributes to the difficulty in accurately predicting their heights.

Solutions to Challenges 1 and 2: Given that the aforementioned challenges originate from the characteristics of the Google satellite image data itself, incorporating supplementary data presents a potential approach to introduce new information externally. Considering that the aim of our method is to extract building heights using data that are consistently available at low cost, street view data emerge as an apt choice. It shares the attributes of ease of acquisition and regular updates similar to Google Earth imagery. A more compelling reason for selecting street view data is its capacity to significantly circumvent the outlined challenges. Street view imagery is captured at high frequencies, and as per our preliminary research, a standard image from the BDH dataset can correspond to between 50 and 300 street view images (depending on the length and density of the road network). Therefore, street view imagery can provide information on buildings obscured by shadows from various ground-level angles. Moreover, street view imagery does not suffer from image stitching issues, contributing to the stability of the model.

Solutions to Challenge 3: To improve the accuracy of height estimation for super high-rise buildings, potential solutions

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TABLE IX
COMPARATIVE ANALYSIS OF GENERALIZATION RESULTS ACROSS DIFFERENT CITIES

Method	Shanghai		Tianjin		Wuhan		Guangzhou		Chengdu	
	AP	MAE (m)	*AP*	MAE (m)	*AP*	MAE (m)	*AP*	MAE (m)	*AP*	MAE (m)
M_l	-	9.45	-	9.13	-	9.57	-	9.33	-	9.38
M_p	-	9.15	-	8.88	-	9.33	-	9.04	-	9.05
Mask R-CNN	21.78	7.46	22.13	7.15	22.35	7.74	21.59	7.70	21.88	7.77
Cascade Mask R-CNN	32.59	7.25	33.14	6.87	33.25	7.42	33.03	7.26	32.96	7.36
TensorMask	17.88	7.58	18.10	7.33	18.32	7.65	17.83	7.62	17.85	7.71
MaskFormer	31.28	6.83	32.04	6.25	32.21	7.18	31.85	6.82	31.97	6.76
Refactored MaskFormer	30.46	6.56	31.17	5.87	31.38	7.04	31.01	6.45	31.04	6.47
LOFT-FOA	31.65	6.18	32.39	5.76	32.42	6.47	32.10	6.15	32.16	6.11
BDH-Net	33.05	5.68	33.73	5.35	33.76	5.71	33.23	5.63	33.31	5.79

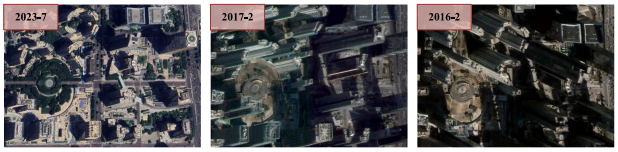


Fig. 21. Variations in off-nadir angles for Google satellite imagery of the same location.

include enhancing the dataset by incorporating more samples of these structures. This can be achieved through oversampling existing instances or generating realistic synthetic data using simulation platforms, thereby expanding and balancing the dataset to better represent super high-rise buildings. Additionally, integrating physical and geometric constraints into the model can address the complex relationship between building height and the roof-to-footprint offset vector, which is more pronounced in super high-rise buildings. These constraints, grounded in structural principles and domain knowledge, can guide the relationship between height and offset, enabling the model to more effectively capture the unique characteristics of these structures and improve prediction accuracy.

B. Generalization Test

For a generalization test, additional experiments were conducted in various cities across China, including Shanghai, Tianjin, Wuhan, Guangzhou, and Chengdu. These cities were chosen to represent different geographical orientations and architectural variations within China. Two sets of imagery were selected from each city for the generalization experiments. The results of building boundary extraction and height estimation are summarized in Table IX. The experimental results indicate that BDH-Net exhibits better generalization capabilities. It is also evident that, due to the geographical proximity and similar architectural styles of Tianjin and Beijing, the results in Tianjin are notably better compared to other cities.

C. Applicability Potential

Although the proposed method is not explicitly tailored for off-nadir imaging, its accuracy diminishes when applied to images with pronounced off-nadir angles. Fig. 21 showcases the Google satellite images of a single location captured at diverse off-nadir angles. The roof-to-footprint offset vector variations are conspicuous. However, Google's provision of a consistent off-nadir angle image series overextended periods, as illustrated in Fig. 22, compensates for this limitation. For example, between 2016 and 2023, nine images exhibit minimal off-nadir angle variations, making the method apt for monitoring building heights over time.

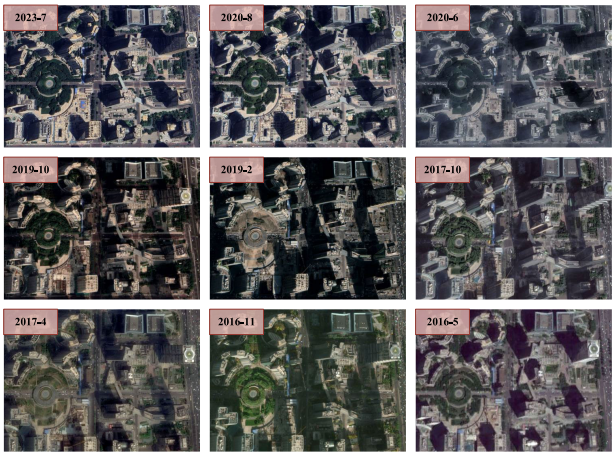


Fig. 22. Consistency in off-nadir angles for Google satellite imagery over time at the same location.

D. Future Work

The Google satellite imagery employed in this research does not inherently furnish information on off-nadir angles and solar elevation angles. This absence impedes two potential enhancements.

1) *Formulating Geometric Relationships for Off-Nadir Angles:* Such equations could steer the deep learning optimization process toward height-relevant features, aiming to improve model interpretability.
2) *Capitalizing on Building Shadow Lengths:* Historically, building shadow length has been an effective proxy for height estimation [44], [91].

Introducing data on off-nadir and solar elevation angles can refine shadow length calculations, potentially reducing issues arising from shadow occlusion in off-nadir imagery. Considering the novelty of this approach, additional investigations are essential to bolster its interpretability, accuracy, and resilience.

VIII. CONCLUSION

This study delineates an approach for extracting 3-D building attributes using monocular off-nadir satellite imagery,

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diverging from conventional methods dependent on orthorectified images and pixel-level semantic segmentation. Our method integrates off-nadir satellite imagery with instance segmentation. We present BDH-Net, an instance-level multitask learning framework that effectively utilizes off-nadir perspectives. This framework effectively capitalizes on roof-to-footprint offset vectors inherent in off-nadir views and models building instance relationships, thereby improving precision in building height estimation. Benefiting from the widespread availability and regular updates of Google Earth imagery, BDH-Net shows the potential for multitemporal applications. This framework aims to enhance the precision and efficiency of 3-D urban modeling while accommodating a variety of applications across different temporal and spatial scales.

The validity of our approach is supported by comprehensive ablation experiments, demonstrating the efficacy of our proposed techniques. This study also validates that the validity of our approach is grounded in a series of comprehensive ablation experiments, demonstrating the efficacy of our proposed techniques. A key aspect of this research is the development of the BDH dataset, compiled from Google Earth imagery. This dataset, with its high spatial resolution of 0.54 m, comprises 931 image samples each of 2048 × 2048 pixels, culminating in more than 67 000 instance-level annotations. It provides an in-depth analysis of the urban landscape within the 667 km^2 area of Beijing's 5th Ring Road. Significantly, the BDH dataset is one of the first specifically curated for the task of building instance segmentation and height estimation from off-nadir satellite sensor imagery.

In conclusion, this research presents a step toward addressing persistent challenges in monocular 3-D building information extraction and contributes a dataset that may facilitate further research in off-nadir building information extraction.

APPENDIX
MASKED CROSS-ATTENTION

Masked cross-attention seamlessly fuses pixel-level feature information with instance-level query features, directing the query's focus toward local features over global ones. For an input query feature $\mathcal{Q}_i^{\text{init}} \in \mathbb{R}^{N_Q \times E_Q}$ from the $(i-1)$th CSA module, alongside a mask feature F_{mask}, and the feature map F_i' at the ith layer, the mechanism of masked cross-attention unfolds as follows.

Initially, mask embeddings $\mathcal{A}_M \in \mathbb{R}^{N_Q \times d_M}$ are generated from $\mathcal{Q}_i$ using an MLP. An attention map $\text{AM}_i \in \mathbb{R}^{N_h \times N_Q \times (HW/16)}$ is then created through the multiplication of the mask feature F_{mask} with the mask embeddings $\mathcal{A}_M$, where N_h indicates the number of attention heads. The attention map AM_i is resized to match the dimensions of F_i' and is then flattened. The masked attention map MAM_i is derived by applying a sigmoid activation to AM_i, followed by thresholding with a standard threshold of 0.5 to produce a binary mask.

This masked cross-attention process computes three crucial components: the query $\mathcal{Q}_i$, the key $\mathcal{K}_i$, and the value $\mathcal{V}_i$, as expressed in (26)–(28). To accommodate the inherent permutation invariance of Transformers, positional embeddings are integrated before calculating the attention weights between the query and the key [92]. Specifically, positional embeddings e_{pos}^q are added to the initial query $\mathcal{Q}_i^{\text{init}}$, and similarly, positional embeddings e_{pos}^f are added to the pixel-level features F_{4-i}' to generate the key $\mathcal{K}_i$. Additionally, a scale-level embedding, e_{level}, is introduced to accurately assign each query pixel to its respective feature layer [79]

$$\mathcal{Q}_i = \mathcal{Q}_i^{\text{init}} + e_{\text{pos}}^q \tag{26}$$

$$\mathcal{K}_i = W_K\left(F_{4-i}' + e_{\text{level}}\right) + e_{\text{pos}}^f \tag{27}$$

$$\mathcal{V}_i = W_V\left(F_{4-i}'\right). \tag{28}$$

This comprehensive approach enriches the query with the masked attention map to building instances, optimizing the focus on relevant local features. The masked query $\mathcal{Q}_i^{\text{mask}}$ is computed as follows:

$$\mathcal{Q}_i^{\text{mask}} = \text{Softmax}\left(\text{Norm}\left(\mathcal{Q}_i\mathcal{K}_i^T \cdot \text{MAM}_i\right)\right)\mathcal{V}_i + \mathcal{Q}_i^{\text{init}}. \tag{29}$$

Subsequently, the masked query $\mathcal{Q}_i^{\text{mask}}$ undergoes self-attention and an FFN, resulting in the updated query $\mathcal{Q}_{i+1}^{\text{init}}$.

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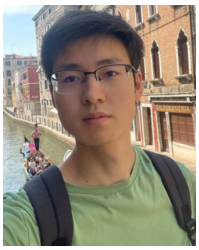
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